

DYNAMICAL SYSTEMS

PROBLEM SET 1

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We begin by noting that all of the files referenced in this document are available at:
<https://github.com/chtompki/MathFall2026/tree/main/DynamicalSystems/ProblemSet1>

Problem Set Instructions:

- In your first attempt of the problem set problems, you are encouraged to treat the problem set as an open-notes quiz. Work through it on your own, without consulting other people, internet resources, or any solutions/answers. Work on each problem, completing as much as you are able to, and making a note in your work whenever you become stuck or confused.
- After your initial individual attempt, collaboration is encouraged as you continue to work on the problems. You can discuss the problem and potential approaches with your classmates, and use resources from class or other sources/references (including related journal articles). However, please do not look at the student solution manual or other posted solutions until after you have submitted your work in Gradescope.
- In your submission, please show all your work and include any code you create in addition the output of your computations. Be sure to acknowledge any substantial help you receive. This includes collaboration with others, references and online resources.

Part 1:

1. Become a student member of the Society for Industrial and Applied Mathematics (SIAM) (You don't need to turn anything in). Because VCU has a SIAM Student Chapter, you can become a member for free.

- (1) Find out more details and sign up at this link:

<http://siam.org/membership/individual-membership/student-membership/>

- (2) Look on the SIAM website for resources/events that might be of interest to you. Also take a look in the SIAM News section linked below for any interesting articles related to applications of dynamical systems:

<https://www.siam.org/publications/siam-news/>

2. Solving ODEs and phase portrait analysis with Matlab. You don't need to turn anything in for this, but use what you've learned to complete the computational aspects of the questions below. You can also feel free to use Python, Mathematica, or other software/programming languages instead of Matlab for your problem sets if you prefer. If you do want to use something different, please discuss this with me beforehand so that we can make sure you will still be able to do everything required over the course of the semester.

- (1) Get access to Matlab if you don't already have it. Information about downloading Matlab or using the online version as a VCU student can be found at the link below:
<https://knowledgebase.vcu.edu/portal/app/portlets/results/viewSolution.jsp?solutionid=250303090503713>
- (2) if you are unfamiliar with Matlab, go through the Matlab onramp tutorial linked below:
<https://matlabacademy.mathworks.com/details/matlab-onramp/gettingstartedL>
- (3) Complete the "SlopeFields.mlx" and "Equilibria1D.mlx" tutorials listed at the link below. In addition to reviewing some concepts related to one-dimensional dynamical systems, these tutorials provide example code for numerically analyzing 1D systems in Matlab. You may either download them or run them using Matlab Online. Link to Tutorials Main Page: <https://www.mathworks.com/matlabcentral/fileexchange/95513-qualitative-analysis-of-odes>
- (4) Complete the following tutorial for using Matlab to implement the Explicit Euler Method for solving and initial value problem:
<https://www.mathworks.com/videos/solving-odes-in-matlab-1-euler-ode1-117526.html>

3. Computing bifurcation diagrams by numerical continuation in COCO. The numerical continuation package is written in Matlab, and will be useful for computing bifurcation diagrams. There are other options for those who prefer alternatives to Matlab, so please talk to me if that is the case.

- (1) download and install COCO using the instructions at the link below:
<https://sourceforge.net/p/cocotools/wiki/Home/>
- (2) Go through Sections 1–3.3 of the "getting started" tutorial, along with links to the related Matlab tutorial file, which are both linked below (You don't need to turn anything in). Note that both the pdf and matlab file linked below are also included in your COCO installation directory.
 Tutorial: https://drive.google.com/file/d/1I0Wm5p9UDiUEiFQ2IRPoUyvXF_N2w8D4/view?usp=sharing
 Matlab Files: https://drive.google.com/file/d/1p6eaDWMDerjFs7B1WBEj00ZrdsKQk-kK/view?usp=drive_link

Part 2: from Strogatz, 2nd Edition, Chapter 2

2.2.4 Analyze the following equation graphically. Sketch the vector field on the real line find all the fixed points, classify their stability, and sketch the graph of $x(t)$ for different initial conditions. Then try to find the analytical solution for $x(t)$, or state that it's impossible. The equation in question is:

$$\dot{x} = e^{-x} \sin(x)$$

Solution. Set

$$f(x) = e^{-x} \sin x.$$

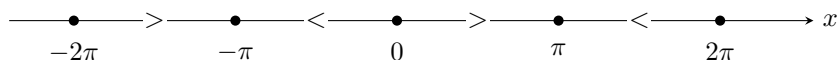
Because $e^{-x} > 0$ for every $x \in \mathbb{R}$, the sign of $f(x)$ is the same as the sign of $\sin x$. The fixed points are therefore

$$f(x_*) = 0 \iff \sin x_* = 0 \iff \boxed{x_* = n\pi, \quad n \in \mathbb{Z}.}$$

Moreover,

$$\dot{x} > 0 \quad \text{on } (2k\pi, (2k+1)\pi), \quad \dot{x} < 0 \quad \text{on } ((2k-1)\pi, 2k\pi).$$

Thus a representative portion of the phase line is



For the stability classification, differentiate:

$$f'(x) = e^{-x}(\cos x - \sin x).$$

At a fixed point $x_* = n\pi$,

$$f'(n\pi) = (-1)^n e^{-n\pi}.$$

Consequently,

$$x = 2k\pi \text{ is unstable, } \quad x = (2k+1)\pi \text{ is asymptotically stable.}$$

All the fixed points are hyperbolic.

Now let $x(0) = x_0$. If $x_0 = n\pi$, uniqueness gives the equilibrium solution

$$x(t) \equiv n\pi.$$

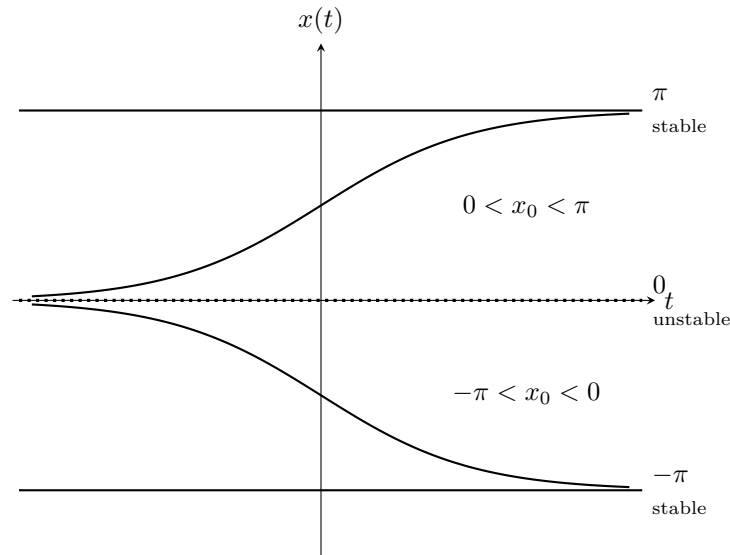
If $2k\pi < x_0 < (2k+1)\pi$, then $x(t)$ increases monotonically and

$$\lim_{t \rightarrow -\infty} x(t) = 2k\pi, \quad \lim_{t \rightarrow \infty} x(t) = (2k+1)\pi.$$

If $(2k-1)\pi < x_0 < 2k\pi$, then $x(t)$ decreases monotonically and

$$\lim_{t \rightarrow -\infty} x(t) = 2k\pi, \quad \lim_{t \rightarrow \infty} x(t) = (2k-1)\pi.$$

Thus every nonconstant orbit connects an unstable even multiple of π to one of the adjacent stable odd multiples. Representative solution curves near $x = 0$ are shown below.



The drawn curves are qualitative sketches; they indicate the correct monotonicity and limiting equilibria, not an explicit formula for the solutions.

For a nonconstant solution, separation of variables yields

$$\frac{e^x}{\sin x} dx = dt.$$

Hence, as long as $x(t)$ remains in the interval between the neighboring fixed points containing x_0 ,

$$t = \int_{x_0}^{x(t)} \frac{e^u}{\sin u} du.$$

The antiderivative $\int e^x \csc x \, dx$ is not an elementary function, so there is no elementary closed-form expression for $x(t)$. If

$$F(x) = \int^x \frac{e^u}{\sin u} \, du$$

is defined on the interval containing x_0 , then the solution may be written implicitly, or in inverse-function form, as

$$F(x(t)) - F(x_0) = t, \quad \boxed{x(t) = F^{-1}(t + F(x_0))}.$$

Together with the constant solutions $x(t) \equiv n\pi$, this describes all solutions. $\square$

2.2.7 Analyze the following equation graphically. Sketch the vector field on the real line find all the fixed points, classify their stability, and sketch the graph of $x(t)$ for different initial conditions. Then try to find the analytical solution for $x(t)$, or state that it's impossible. The equation in question is:

$$\dot{x} = e^x - \cos(x)$$

(Hint: Sketch the graphs of e^x and $\cos(x)$ on the same axes, and look for intersections. You won't be able to find the fixed points explicitly, but you still can find the qualitative behaviour.)

Solution. Set

$$f(x) = e^x - \cos x.$$

Fixed points are intersections of $y = e^x$ and $y = \cos x$. For $x > 0$, $e^x > 1 \geq \cos x$, while $e^0 = \cos 0 = 1$. Hence $x = 0$ is the only nonnegative fixed point.

For $x < 0$, an intersection is possible only where $\cos x > 0$. On every positive cosine lobe centered at $-2n\pi$, $n \geq 1$, one has

$$f\left(-\left(2n + \frac{1}{2}\right)\pi\right) > 0, \quad f(-2n\pi) = e^{-2n\pi} - 1 < 0,$$

and

$$f\left(-\left(2n - \frac{1}{2}\right)\pi\right) > 0.$$

Therefore there are two fixed points on each such lobe. There is also one fixed point in $(-\pi/2, 0)$, in addition to the fixed point at 0.

Write the negative fixed points as

$$a_0 \in \left(-\frac{\pi}{2}, 0\right)$$

and, for $n \geq 1$,

$$a_n \in \left(-\left(2n + \frac{1}{2}\right)\pi, -2n\pi\right), \quad b_n \in \left(-2n\pi, -\left(2n - \frac{1}{2}\right)\pi\right).$$

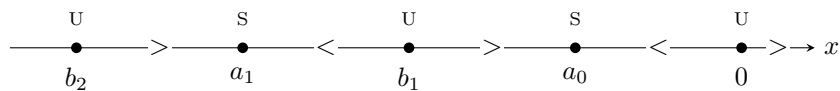
Thus all the fixed points are

$$\boxed{0, \quad a_0, \quad a_n, \quad b_n \quad (n \geq 1)}.$$

They cannot be expressed in elementary closed form. Numerically, the first few are

$$a_0 \approx -1.29270, \quad b_1 \approx -4.72129, \quad a_1 \approx -7.85359, \quad b_2 \approx -10.99559.$$

As one moves from left to right across a positive cosine lobe, the sign of f changes from positive to negative at a_n , and from negative to positive at b_n . The same positive-to-negative change occurs at a_0 , whereas f changes from negative to positive at 0. A representative portion of the phase line is therefore



Here S denotes asymptotically stable and U denotes unstable. Consequently,

$$a_n \text{ is asymptotically stable for every } n \geq 0,$$

whereas

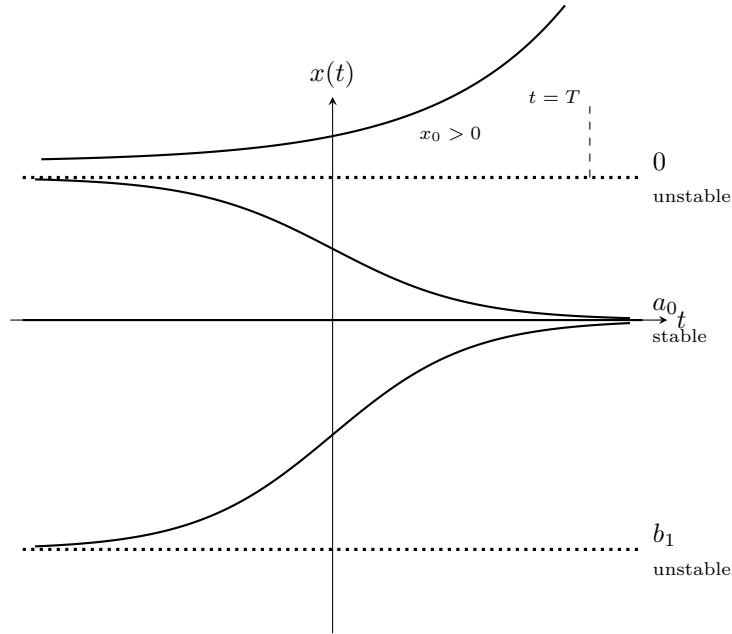
$$b_n \text{ is unstable for every } n \geq 1, \text{ and } x = 0 \text{ is unstable.}$$

In particular, $f'(0) = e^0 + \sin 0 = 1 > 0$, confirming directly that 0 is unstable.

The phase line determines all the solution behavior. Between any unstable fixed point and the stable fixed point immediately to its right, solutions increase toward that stable point. Between a stable fixed point and the unstable fixed point immediately to its right, solutions decrease toward the stable point. For example,

$$\begin{aligned} b_1 < x_0 < a_0 : \quad x(t) \nearrow a_0 \quad (t \rightarrow \infty), \quad x(t) \rightarrow b_1 \quad (t \rightarrow -\infty), \\ a_0 < x_0 < 0 : \quad x(t) \searrow a_0 \quad (t \rightarrow \infty), \quad x(t) \rightarrow 0 \quad (t \rightarrow -\infty). \end{aligned}$$

Initial data at a fixed point give a constant solution. Representative solution curves are shown below; solid horizontal lines are stable equilibria and dotted horizontal lines are unstable equilibria.



The curves are qualitative sketches rather than exact solution formulas. For $x_0 > 0$, $f(x) > 0$, so the solution increases without bound. In fact it reaches $+\infty$ in finite forward time because, for sufficiently large x ,

$$e^x - \cos x \geq \frac{1}{2}e^x, \quad \int^\infty \frac{dx}{e^x - \cos x} < \infty.$$

Thus $x(t) \rightarrow +\infty$ as $t \uparrow T$ for some finite T .

Finally, every nonconstant solution is separable:

$$\frac{dx}{e^x - \cos x} = dt.$$

With $x(0) = x_0$, its exact implicit form is

$$t = \int_{x_0}^{x(t)} \frac{du}{e^u - \cos u}.$$

The antiderivative is not elementary, so no elementary explicit formula for $x(t)$ is available. On any interval that contains no fixed point, define

$$F(x) = \int \frac{du}{e^u - \cos u}.$$

Then the solution can equivalently be written in inverse-function form as

$$\boxed{x(t) = F^{-1}(t + F(x_0)).}$$

Together with the constant solutions at all the fixed points, this describes all solutions. $\square$

2.2.8 (Working backwards, from flows to equations) Given an equation

$$\dot{x} = f(x),$$

we know how to sketch the corresponding flow on the real line. Here you are asked to solve the opposite problem: For the phase portrait shown in Figure 1, find an equation that is consistent with it. (There are an infinite number of correct answers—and wrong ones too.)

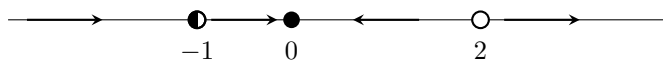


FIGURE 1.

Solution. The fixed points in the portrait are

$$x = -1, \quad x = 0, \quad \text{and} \quad x = 2.$$

The arrows prescribe the following sign pattern for f :

x	$(-\infty, -1)$	$(-1, 0)$	$(0, 2)$	$(2, \infty)$
$f(x)$	+	+	−	+

At $x = -1$, the sign does not change. Thus the simplest polynomial should have a zero of even multiplicity there; we choose the factor $(x + 1)^2$. At both $x = 0$ and $x = 2$, the sign changes, so we choose simple factors x and $(x - 2)$. This suggests

$$f(x) = (x + 1)^2 x (x - 2).$$

Indeed, $(x + 1)^2 \geq 0$, so away from $x = -1$ the sign of f is determined by $x(x - 2)$. Therefore

$$x < 0: \quad x(x - 2) > 0,$$

$$0 < x < 2: \quad x(x - 2) < 0,$$

$$x > 2: \quad x(x - 2) > 0.$$

This gives precisely the required arrows on every interval, including the positive sign on both sides of $x = -1$. Hence one equation consistent with the portrait is

$$\boxed{\dot{x} = (x + 1)^2 x (x - 2).}$$

The stability shown by this equation also agrees with the portrait:

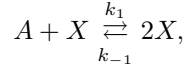
- At $x = -1$, solutions move toward the fixed point from the left and away from it on the right. Thus $x = -1$ is half-stable (stable from the left and unstable from the right).
- At $x = 0$, arrows point toward the fixed point from both sides, so $x = 0$ is asymptotically stable.
- At $x = 2$, arrows point away from the fixed point on both sides, so $x = 2$ is unstable.

The answer is not unique. More generally, for any continuous function $g(x) > 0$ on $\mathbb{R}$,

$$\dot{x} = g(x)(x+1)^2x(x-2)$$

has the same fixed points and the same phase portrait. One could also use higher odd multiplicities at 0 or 2, or a higher even multiplicity at -1 , provided no additional zeros are introduced and the required sign pattern is preserved. $\square$

2.3.2 (Autocatalysis) Consider the model chemical reaction



in which one molecule of X combines with one molecule of A to form two molecules of X . This means that the chemical X stimulates its own production, a process called *autocatalysis*. This positive feedback process leads to a chain reaction, which eventually is limited by a “back reaction” in which $2X$ returns to $A + X$.

According to the *law of mass action* of chemical kinetics, the rate of an elementary reaction is proportional to the product of the concentrations of the reactants. We denote the concentrations by lowercase letters $x = [X]$ and $a = [A]$. Assume that there is an enormous surplus of chemical A , so that its concentration a can be regarded as constant. Then the equation for the kinetics of x is

$$\dot{x} = k_1ax - k_{-1}x^2,$$

where k_1 and k_{-1} are positive parameters called *rate constants*.

- Find all the fixed points of this equation and classify their stability.
- Sketch the graph of $x(t)$ for various initial values x_0 .

Solution. It is convenient to introduce

$$r = k_1a > 0, \quad K = \frac{k_1a}{k_{-1}} > 0.$$

Then the equation becomes the logistic equation

$$\dot{x} = rx \left(1 - \frac{x}{K}\right).$$

- The fixed points satisfy

$$rx \left(1 - \frac{x}{K}\right) = 0.$$

Therefore

$$x_1^* = 0, \quad x_2^* = K = \frac{k_1a}{k_{-1}}.$$

Let

$$f(x) = k_1ax - k_{-1}x^2.$$

Its derivative is

$$f'(x) = k_1a - 2k_{-1}x.$$

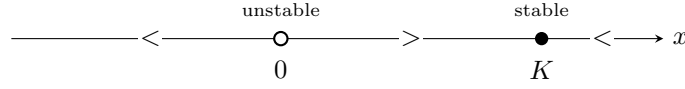
At the fixed points,

$$f'(0) = k_1a > 0, \quad f'(K) = k_1a - 2k_{-1}\frac{k_1a}{k_{-1}} = -k_1a < 0.$$

Hence

$$x = 0 \text{ is unstable,} \quad x = K \text{ is asymptotically stable.}$$

The corresponding phase line is

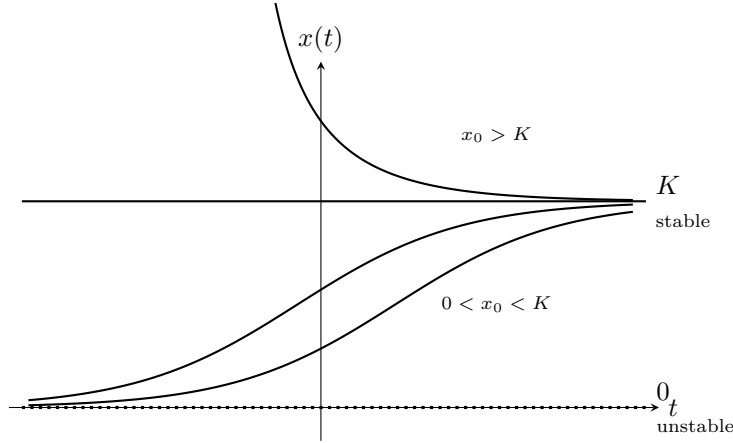


Only $x \geq 0$ is physically meaningful because x is a concentration. On this physical domain, solutions move away from 0 and toward K .

b) Let $x(0) = x_0$. The qualitative behavior is as follows:

Initial value	Behavior as $t \rightarrow \infty$
$x_0 = 0$	$x(t) \equiv 0$,
$0 < x_0 < K$	$x(t)$ increases monotonically to K ,
$x_0 = K$	$x(t) \equiv K$,
$x_0 > K$	$x(t)$ decreases monotonically to K .

Representative solution curves are shown below. The stable equilibrium is solid and the unstable equilibrium is dotted.



For completeness, the equation can also be solved analytically. Separating variables gives

$$\frac{dx}{x(1 - x/K)} = r dt.$$

Since

$$\frac{1}{x(1 - x/K)} = \frac{1}{x} + \frac{1}{K - x},$$

integration yields

$$\ln \left| \frac{x}{K - x} \right| = rt + C.$$

For $x_0 \neq 0$, solving for $x(t)$ gives

$$x(t) = \frac{K}{1 + \left(\frac{K}{x_0} - 1 \right) e^{-rt}} = \frac{\frac{k_1 a}{k_{-1}}}{1 + \left(\frac{k_1 a}{k_{-1} x_0} - 1 \right) e^{-k_1 a t}}.$$

The case $x_0 = 0$ must be included separately and gives $x(t) \equiv 0$.

Mathematically, if $x_0 < 0$, then $\dot{x} < 0$ and the solution decreases toward $-\infty$ in finite forward time. Such solutions are not chemically meaningful, since a concentration cannot be negative.

□

2.3.3 (Tumor growth) The growth of cancerous tumors can be modeled by the Gompertz law

$$\dot{N} = -aN \ln(bN),$$

where $N(t)$ is proportional to the number of cells in the tumor, and $a, b > 0$ are parameters.

- Interpret a and b biologically.
- Sketch the vector field and graph $N(t)$ for various initial values.

The predictions of this simple model agree surprisingly well with data on tumor growth, as long as N is not too small; see Aroesty et al. (1973) and Newton (1980) for examples.

Solution. Because N represents tumor size, the physically meaningful domain is $N \geq 0$. The displayed differential equation is defined for $N > 0$; since

$$\lim_{N \rightarrow 0^+} N \ln(bN) = 0,$$

we may continuously extend its right-hand side by defining $\dot{N} = 0$ at $N = 0$.

- Introduce the limiting tumor size

$$K = \frac{1}{b}.$$

Then the Gompertz law can be written as

$$\dot{N} = aN \ln\left(\frac{K}{N}\right).$$

The parameter b therefore determines the saturation size, or carrying capacity,

$$N_{\infty} = K = \frac{1}{b}.$$

Thus increasing b decreases the predicted limiting tumor size.

The per-capita growth rate is

$$\frac{\dot{N}}{N} = a \ln\left(\frac{K}{N}\right).$$

Hence a sets the time scale on which the tumor approaches its limiting size. In fact, as shown below, the logarithmic distance from the limiting size decays like e^{-at} , so the characteristic time is

$$\tau = \frac{1}{a}.$$

Larger a means faster growth and faster relaxation toward K .

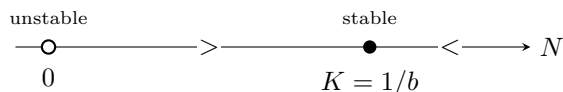
- For $N > 0$, let

$$f(N) = -aN \ln(bN) = aN \ln\left(\frac{K}{N}\right).$$

The fixed points on the extended physical domain are

$$N_1^* = 0, \quad N_2^* = K = \frac{1}{b}.$$

For $0 < N < K$, one has $\ln(K/N) > 0$, so $\dot{N} > 0$. For $N > K$, $\ln(K/N) < 0$, so $\dot{N} < 0$. The phase line is therefore

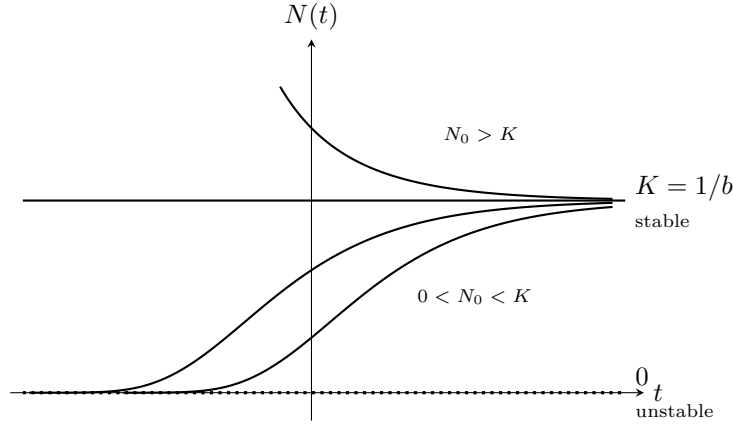


Thus $N = 0$ is unstable from the physically relevant side, while $N = K$ is asymptotically stable. In particular,

$0 < N_0 < K$	: $N(t)$ increases toward K ,
$N_0 = K$	: $N(t) \equiv K$,
$N_0 > K$	: $N(t)$ decreases toward K .

Also, under the continuous extension at the origin, $N_0 = 0$ gives $N(t) \equiv 0$.

Representative solution curves are shown below. The unstable equilibrium is dotted and the stable equilibrium is solid.



The solution can be found explicitly. Set

$$y(t) = \ln(bN(t)).$$

For $N(t) > 0$,

$$\dot{y} = \frac{\dot{N}}{N} = -a \ln(bN) = -ay.$$

Therefore

$$y(t) = y(0)e^{-at} = \ln(bN_0)e^{-at}.$$

Exponentiating gives

$N(t) = \frac{1}{b} \exp(\ln(bN_0)e^{-at}), \quad N_0 > 0.$

Equivalently,

$$N(t) = K \exp\left(\ln\left(\frac{N_0}{K}\right)e^{-at}\right).$$

This formula shows directly that $N(t) \rightarrow K = 1/b$ as $t \rightarrow \infty$ for every $N_0 > 0$.

For a growing solution, the curvature changes sign when $N = K/e$. Thus a solution starting below K/e is initially concave upward and later concave downward, giving the characteristic sigmoidal Gompertz growth curve. If $K/e \leq N_0 < K$, it is concave downward for all $t \geq 0$.

□

2.3.5 (Dominance of the fittest) Suppose X and Y are two species that reproduce exponentially fast: $\dot{X} = aX$ and $\dot{Y} = bY$, respectively, with initial conditions $X_0, Y_0 > 0$ and growth rates $a \geq b \geq 0$. Here X is “fitter” than Y in the sense that it reproduces faster, as reflected by the inequality $a > b$. So we’d expect X to keep

increasing its share of the total population $X + Y$ as $t \rightarrow \infty$. The goal of this exercise is to demonstrate this intuitive result, first analytically and then geometrically.

- (a) Let $x(t) = X(t)/[X(t) + Y(t)]$ denote X 's share of the total population. By solving for $X(t)$ and $Y(t)$, show that $x(t)$ increasing monotonically approaches 1 as $t \rightarrow \infty$.
- (b) Alternatively, we can arrive at the same conclusions by deriving a differential equation for $x(t)$. To do so, take the time derivative of $x(t) = X(t)/[X(t) + Y(t)]$ using the quotient and chain rules. Then substitute for X and Y and thereby show that $x(t)$ obeys the logistic equation $\dot{x} = (a - b)x(1 - x)$. Explain why this implies that $x(t)$ increases monotonically and approaches 1 as $t \rightarrow \infty$.

Solution.

□

2.4.3 Use linear stability analysis to classify the fixed points of the following system. If linear stability analysis fails because $f'(x^*) = 0$, use a graphical argument to decide the stability.

$$\dot{x} = \tan(x)$$

Solution.

□

2.4.8 using linear stability analysis, classify the fixed points of the Gompertz model of tumor growth $\dot{N} = -aN \ln(bN)$. (Similar to Exercise 2.3.3, $N(t)$ is proportional to the number of cells in the tumor and $a, b > 0$ are parameters.)

Solution.

□

2.8.2 Sketch the slope field for the following differential equations. Then “integrate” the equation manually by drawing trajectories that are everywhere parallel to the local slope.

- (a) $\dot{x} = x$
- (c) $\dot{x} = 1 - 4x(1 - x)$

Solution.

□

2.8.6 (Analytically intractable problem) Consider the initial value problem $\dot{x} = x + e^{-x}$, $x(0) = 0$.

- (a) Sketch the solution $x(t)$ for $t \geq 0$
- (b) Using some analytical arguments, obtain rigorous bounds on the value of x at $t = 1$. In other words prove that $a < x(1) < b$, for a, b to be determined. By being clever, try to make a and b as close together as possible. (Hint: Bound the given vector field by approximate vector fields that can be integrated analytically.)
- (c) Now for the numerical part: Using the Euler method, compute x at $t = 1$, correct to three decimal places. How small does the stepsize need to be to obtain the desired accuracy? (Give the order of magnitude, not the exact number)

Solution.

□